Bittle X Platform v1.0 - Interface Specification

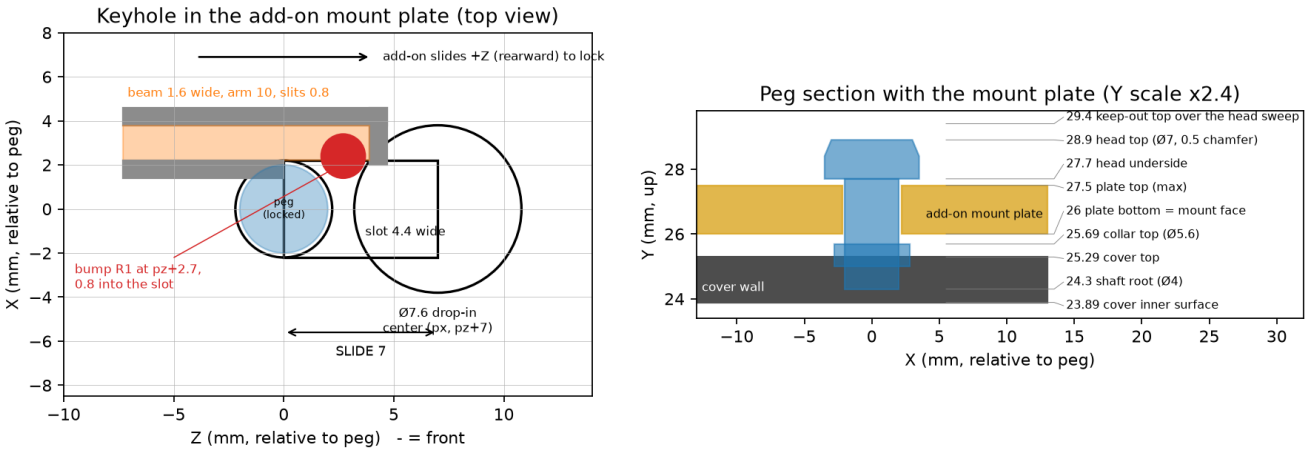
Version 1.0, 2026-10-14. The platform is the stock Petoi Bittle X back cover with 4 pegs on its top. An add-on has a plate with 4 keyholes. Put it on the pegs, slide it 7 mm rearward, and it clicks. No tools, glue or tape are necessary. interface.json has all the numbers in machine-readable form.



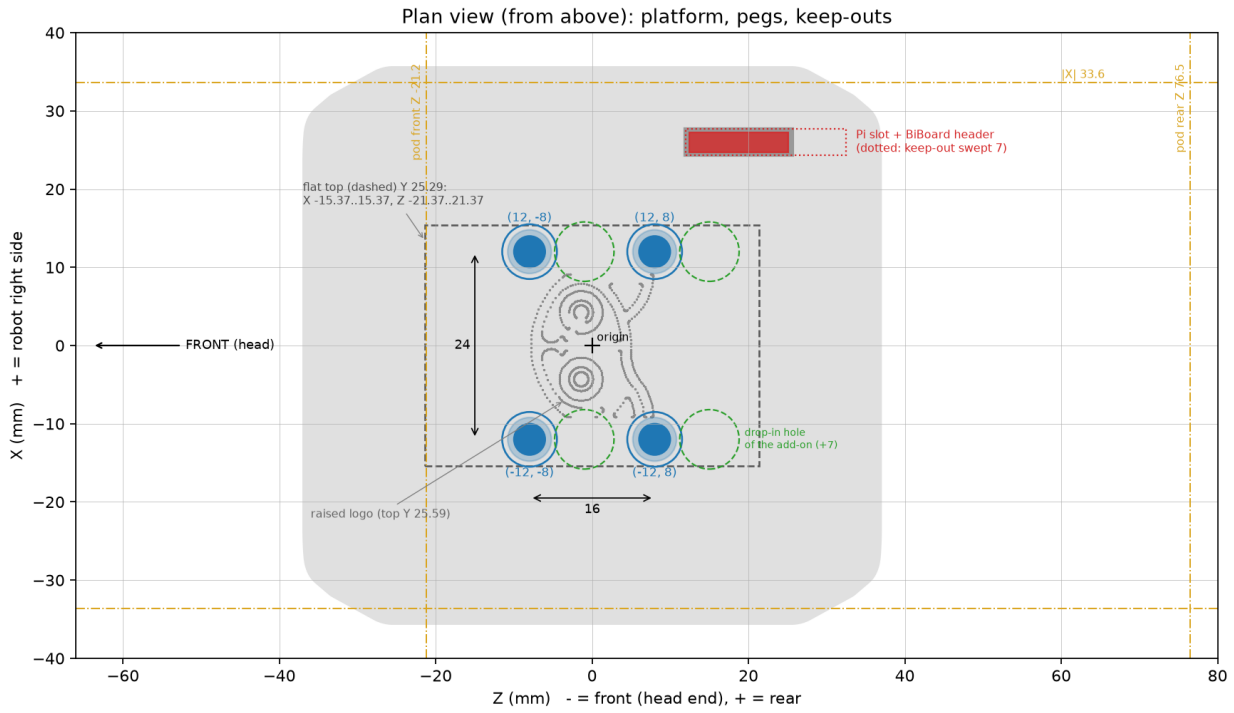
The platform (dark) with the starter plate (yellow).

1. Key numbers

Item	Value (mm)
Frame	X lateral (+X = robot right), Y up, Z along the body (-Z = head end). Origin: center line, middle of the cover.
Pegs (X, Z)	(12, -8), (12, 8), (-12, -8), (-12, 8): a centered 24 x 16 rectangle
Peg	shaft Ø4; head Ø7, Y 27.7 to 28.9
Add-on plate	bottom at Y 26, 1.5 thick at the keyholes
Keyhole	slot 4.4 wide from the peg to 7 mm behind it (+Z); Ø7.6 drop-in hole at the rear end
Fit	put the add-on on 7 mm forward of its place, then slide it 7 mm rearward until it clicks
Click (detent)	a 1.6 wide spring beam on the +X side of each slot, freed by a 0.8 U slit (fixed at its front end only); bump R1, 0.8 into the slot, 2.7 behind the peg
Keep clear	the peg heads as they slide; the BiBoard header in the cover slot (X 24.73..27.27, Z 12.4..25.1, top Y 29.1)
Size	the pod fits in Z -21.2..76.5, |X| <= 33.6; outside that, check the legs and the head



Left: keyhole and click beam in the add-on plate (top view). Right: the peg and the plate (side section).



Plan view: pegs (blue), drop-in holes (green), BiBoard header (red), pod size (gold).

2. Design an add-on

1 Start from cad/starter_plate.step. It has the keyholes and the click beams.

2 Add your part on top of the plate, in the same print. You can also screw parts to the M3 holes.

3 Keep your part out of stl/installed/keepout.stl. This keeps it clear of the robot, the peg heads and the header.

4 Keep the weight low and near the center line. The pegs carry the whole add-on.

5 For a plate of a different shape: subtract cad/keyhole_cutter.step, then add cad/detent_bumps.step. Both are in position.

3. Printing

Part	Orientation	Supports	Cura profile
cover_pegs	stand on the rear end	tree, from the bed	Bittle Cover - PLA 0.25 (14-line brim)
starter_plate and add-ons	plate bottom on the bed	none	Bittle Pi Pod - PLA 0.25
small, detailed parts	-	-	Bittle Fine - PLA 0.10

Setting	0.25 profiles	Fine 0.10
Layer	0.25	0.10 (first layer 0.25)
Walls / infill	3 / 20 % gyroid	3 / 20 % gyroid
Outer wall / inner wall / infill speed	30 / 40 / 60 mm/s	25 / 35 / 50 mm/s
Outer wall acceleration	600 mm/s²	500 mm/s²
Nozzle	220 C (PLA)	220 C (PLA)

1 Material: PLA works. Use PETG for add-ons that you remove often or that get warm.

2 Standing on its end, the cover prints the pegs with layers along their length. This makes them strong.

3 In add-ons, keep overhangs at 45 deg or less. Use chamfers, not round edges, on the bottom side.

4 The fits use 0.2 to 0.3 clearance. Calibrate your printer if the plate does not slide on.

4. Files

File	Content
stl/print/	the parts to print, in print orientation

File	Content
cad/starter_plate.step	the start point for an add-on
cad/pegs.step, keyhole_cutter.step, detent_bumps.step	the interface parts, in position (robot frame)
stl/installed/	all parts and keepout.stl in the robot frame
interface.json / interface.py	all numbers
build.py, docs.py	rebuild the parts and this document (build123d, manifold3d)

Open items: the BiBoard header height is an estimate from photos. The legs, the head and the robot balance are not modeled. The Petoi cover is a mesh, so cover_pegs is STL only.

5. License

The designs and this document are CC BY 4.0, and the code is MIT (Jeremy Collins). The cover, the side shoulder and the front plate are Petoi files from github.com/PetoiCamp/NonCodeFiles. Petoi publishes them without a stated license, and cover_pegs.stl is made from the Petoi cover. LICENSE.md and ATTRIBUTION.md in the repository give the details.